VA

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Projects

FairGavel - Fairness-Driven Legal Judgment Prediction with GNNs
VIT Chennai • December 2024 - April 2025

- Led a capstone project to build a **fairness-aware legal judgment prediction system** using GNNs
- Designed **graph construction pipeline** using **BERT** for sentence nodes, legal entity linking, and multi-type edges
- Developed the core training architecture consisting of **GraphSAGE** encoder with adversarial debiasing via **GRL** and **attention pooling**
- Focused on modeling fairness for **sensitive attributes** while maintaining predictive accuracy

Non-Intrusive Stress Prediction System using Sleep Position Analysis and Infrared Imaging
VIT Chennai • August 2023 - December 2024

- Acquired dataset with mannequin-based **infrared** images under low-light from an **IR-selective camera**
- Used **YOLO** to detect and normalize 17 body **landmarks** for pose analysis
- Extracted **Euclidean-distance, angular, symmetry and spread** features; extensively tested multiple approaches to derive the **optimal ensemble**
- Fused images into a **Landmark Fusion** model and fed poses into a regression model for **non-intrusive, secure, real-time** stress estimation

Real-Time Multimodal Sign Language Detection
VIT Chennai • December 2023 - April 2024

- Built a **real-time ASL translation** prototype on **Raspberry Pi**, combining hand-gesture, facial-expression and posture models
- Designed and optimized a **compact multimodal model** to run efficiently on Raspberry Pi without sacrificing translation quality
- Engineered an ESP32 → Raspberry Pi wireless pipeline: **ESP32** streams video; Pi runs **MediaPipe** based model for ASL detection
- Integrated an **RPi-display** for live text output and scoped future **smart-glass/watch** wearable integration

TextToTune - Lyric-Based Melody Generation
VIT Chennai • December 2023 - April 2024

- Leveraged a 12K-song **MIDI–lyrics dataset** with extracted syllable–note alignments
- Designed and implemented a conditional **LSTM-GAN** with deep LSTM generator & discriminator both conditioned on input lyrics
- Simultaneously generated melody and alignment **mapping from syllables** to notes in an end-to-end framework
- Produced tuneful, coherent sequences that align closely with lyrical phrasing

Education

08/2025
Velore Institute of Technology Chennai
Chennai, India
Bachelor of Technology: CSE With Specialization in AI&ML

07/2021
SBOA School And Junior College
Chennai, India
AISSCE

Certificates

TOEFL
ETS • 2025

Google Cloud Digital Leader
Google CLoud • 2023

AWS Cloud Practitioner
AWS • 2023

Entrepreneurship
NPTEL • 2023

Ethical Hacking
NPTEL • 2022

C
Advanced CPP
PHP and MySQL
Java
Python
Spoken Tutorial • 2021

Legal Text Summarizer using GCNs
VIT Chennai • June 2024 - December 2024

- Tokenized legal documents with **BERT** and built a **graph** of sentences/entities as nodes with multi-type edges
- Processed the graph through **multi-layer GCNs** to capture inter-clause dependencies and contextual relations
- Fused GCN-derived embeddings with a **transformer encoder–decoder** for coherent, concise summary generation
- Delivered **context-rich summaries** that improve interpretability of dense legal texts and support faster legal research

Animal Intrusion Detection with Deep Learning
VIT Chennai • April 2023 - July 2023

- **Collected, preprocessed and labeled** the data-set of 2000 animals using Python.
- **Identified and separated** at least 3 different unique animals in a video using deep learning and Python.
- Implemented the **deep learning algorithm in a web app** using the **Streamlit** framework to detect animals with high accuracy.
- Upon intrusion detection, an **alert message gets dispatched** to the user's mobile phone. Opportunities for further advancement exist in the areas of animal localization and tracking.

Cyber Combat - A 2D SHooter Game
SBOA School and Junior College • January 2020 - April 2021

- Created a **Horizontal Scrolling 2D game** with **Python** with 6 levels and 3 hours worth of gameplay
- Programmed **4 different enemies** with unique attacking patterns
- Implemented **Character Control** and different mechanics for **2 different weapons**
- Added a **Main Menu, Pause Screen and Levels and Saving mechanics** for maximum User Friendliness

Involvement

VIT Chennai - Game Developer
Chennai
05/2022 - Current

- Developed a 3D game using Unreal Engine, Blueprints and C++
- Increased user retention by creating visually appealing and immersive game environments and enjoyable gameplay
- Managed a Team of 5 members
- Published the game in my College Cultural Fest - Vibrance
- Participated in some marketing and management during my College Cultural Fest - Vibrance

VIT Chennai - Organizer
Chennai, India
02/2023 - 03/2023

- Managed and Lead the Food and Shops committee during the 3 day college cultural event Vibrance
- Streamlined event logistics by coordinating with vendors, managing budgets, and creating detailed timelines
- Managed both the main Food Court and the mini shops during pro shows and concerts on all 3 days

Honors & Awards

First Place - Inter House Cultural Competition
SBOA JC • 2020
Third Place - Inter-School Web Development
Myriad Hues • 2019
First Place - Academic Proficiency
SBOA JC • 2011, 2012, 2013, 2014

Skills

Programming Languages

- C / C++/C#/C Sharp
- Python
- Java
- Unity/Unreal Engine
- HTML/CSS/Javascript/Django
- SQL/NoSQL
- Assembly

Domains

- Game Development
- App Development
- Image Processing
- Graphical Neural Networks
- Machine Learning and Deep Learning
- NLP
- DSA/DAA
- Web Development
- IoT, Embedded Systems
- AWS/Google Cloud Platform
- Arduino, Raspberry PI, ESP32

Soft Skills

- Collaboration and Communication
- Leadership
- Organizational
- Innovation
- Problem-Solving

Languages

English
Bilingual or Proficient (C2)

Hindi
Bilingual or Proficient (C2)

Tamil
Bilingual or Proficient (C2)

French
Elementary (A2)